THE LOCALIZED CHERN CLASSES OF THE STRUCTURE SHEAF OF A REGULAR CLOSED IMMERSION VANISH BELOW THE CODIMENSION AND EQUAL A FACTORIAL MULTIPLE OF THE CLASS OF THE NORMAL SHEAF IN TOP DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FEG (sources/stacks-chow/pinned-original-source.tex line 12613).

Allowed standard prerequisites (editorial).

- Localized Chern classes of a perfect object and the bivariant class of a locally free sheaf. Same chapter, Situation labelled situation-loc-chern and Definition labelled definition-localized-chern, invoked in the statement; Lemma labelled lemma-independent-loc-chern-bQ, cited at proof lines 12656-12658 and 12674-12676. Supplies the classes being computed and the defining property that converts the computation into one about the restriction to E.
- Regular immersions, Koszul resolutions and the perfection of the structure sheaf. Divisors, Definition labelled divisors-definition-regular-immersion and Lemma labelled divisors-lemma-Noetherian-scheme-regular-ideal, cited at proof lines 12634-12637; More on Algebra, Lemma labelled more-algebra-lemma-regular-koszul-regular, cited at proof lines 12638-12640. Realises the structure sheaf of Z as a perfect object with the required vanishing off Z.
- The blowing up in infinity(Z) and its affine charts as complete intersections. Same chapter, Section labelled section-blowup-Z-first, invoked at proof lines 12650-12652; More on Algebra, Lemma labelled more-algebra-lemma-blowup-regular-sequence, cited at proof lines 12688-12690. Provides the morphism b and the chart computation showing that i' is a regular immersion, hence Q is perfect.
- The vanishing and top-degree computation for the fiber case. Same chapter, Lemma labelled lemma-compute-section, cited at proof lines 12676-12678

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and 12702-12705. Supplies both the vanishing of c_t for t < r and the top-degree formula that is transported back.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 0FB5: sources/stacks-chow/pinned-original-source.tex, source lines 10787–10787. Same-chapter pointer into the author environment definition at source line 10787; invoked at proof line 12646 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 063J: sources/stacks-chow/tags.txt, source lines 7833–7833. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12634 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 063I: sources/stacks-chow/tags.txt, source lines 7832–7832. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12635 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 062F: sources/stacks-chow/tags.txt, source lines 7794–7794. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12638 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0GUJ: sources/stacks-chow/pinned-original-source.tex, source lines 10678–10678. Same-chapter pointer into the author environment situation at source line 10678; invoked at proof line 12645 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0FBG: sources/stacks-chow/pinned-original-source.tex, source lines 11646–11646. Same-chapter pointer into the author environment lemma at source line 11646; invoked at proof line 12651 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0FE9: sources/stacks-chow/pinned-original-source.tex, source lines 11678–11678. Same-chapter pointer into the author environment lemma at source line 11678; invoked at proof line 12651 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FE8: sources/stacks-chow/pinned-original-source.tex, source lines 11165–11165. Same-chapter pointer into the author environment lemma at source line 11165; invoked at proof line 12659 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FEF: sources/stacks-chow/pinned-original-source.tex, source lines 12542–12542. Same-chapter pointer into the author environment lemma at source line 12542; invoked at proof line 12671 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9J: sources/stacks-chow/pinned-original-source.tex, source lines 10026–10026. Same-chapter pointer into the author environment lemma at source line 10026; invoked at proof line 12680 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BIQ: sources/stacks-chow/tags.txt, source lines 14498–14498. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12691 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9K: sources/stacks-chow/pinned-original-source.tex, source lines 10204–10204. Same-chapter pointer into the author environment lemma at source line 10204; invoked at proof line 12704 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let $i : Z \to X$ be a regular closed immersion of codimension r between schemes locally of finite type over S. Let $\mathcal{N} = \mathcal{C}_{Z/X}^\vee$ be the normal sheaf. If X is quasi-compact (or has quasi-compact irreducible components), then $c_t(Z \to X, i_*\mathcal{O}_Z) = 0$ for $t = 1, \ldots, r-1$ and*

$$c_r(Z \to X, i_*\mathcal{O}_Z) = (-1)^{r-1}(r-1)!c(Z \to X, \mathcal{N}) \quad \text{in} \quad A^r(Z \to X)$$

where $c_t(Z \to X, i_*\mathcal{O}_Z)$ is the localized Chern class of Definition 0FB5.

Proof. For any $x \in Z$ we can choose an affine open neighbourhood $\mathrm{Spec}(A) \subset X$ such that $Z \cap \mathrm{Spec}(A) = V(f_1, \ldots, f_r)$ where $f_1, \ldots, f_r \in A$ is a regular sequence. See Divisors, Definition 063J and Lemma 063I. Then we see that the Koszul complex on $f_1, \ldots, f_r$ is a resolution of $A/(f_1, \ldots, f_r)$ for example by More on Algebra, Lemma 062F. Hence $A/(f_1, \ldots, f_r)$ is perfect as an A-module. It follows that $F = i_*\mathcal{O}_Z$ is a perfect object of $D(\mathcal{O}_X)$ whose restriction to $X \setminus Z$ is zero. The assumption that X is quasi-compact (or has quasi-compact irreducible components) means that the localized Chern classes $c_t(Z \to X, i_*\mathcal{O}_Z)$ are defined, see Situation 0GUJ and Definition 0FB5. All in all we conclude that the statement makes sense.

Denote $b : W \to \mathbf{P}^1_X$ the blowing up in $\infty(Z)$ as in Section 0FBG. By (0FE9) we have a closed immersion

$$i' : \mathbf{P}^1_Z \longrightarrow W$$

We claim that $Q = i'_*\mathcal{O}_{\mathbf{P}^1_Z}$ is a perfect object of $D(\mathcal{O}_W)$ and that F and Q satisfy the assumptions of Lemma 0FE8.

Assume the claim. The output of Lemma 0FE8 is that we have

$$c_p(Z \to X, F) = c'_p(Q) = (E \to Z)_* \circ c'_p(Q|_E) \circ C$$

for all $p \geq 1$. Observe that $Q|_E$ is equal to the pushforward of the structure sheaf of Z via the morphism $Z \to E$ which is the base change of i' by ∞. Thus the vanishing of $c_t(Z \to X, F)$ for $1 \leq t \leq r-1$ by Lemma 0FEF applied to $E \to Z$. Because $\mathcal{C}_{Z/X} = \mathcal{N}^\vee$ is locally free the bivariant class $c(Z \to X, \mathcal{N})$ is characterized by the relation

$$j^* \circ C = p^* \circ c(Z \to X, \mathcal{N})$$

where $j : C_Z X \to W_\infty$ and $p : C_Z X \to Z$ are the given maps. (Recall $C \in A^0(W_\infty \to X)$ is the class of Lemma 0F9J.) Thus the displayed equation in the statement of the lemma follows from the corresponding equation in Lemma 0FEF.

Proof of the claim. Let A and $f_1, \ldots, f_r$ be as above. Consider the affine open $\mathrm{Spec}(A[s]) \subset \mathbf{P}^1_X$ as in Section 0FBG. Recall that $s = 0$ defines $(\mathbf{P}^1_X)_\infty$ over this open. Hence over $\mathrm{Spec}(A[s])$ we are blowing up in the ideal generated by the regular

sequence $s, f_1, \ldots, f_r$. By More on Algebra, Lemma 0BIQ the $r+1$ affine charts are global complete intersections over $A[s]$. The chart corresponding to the affine blowup algebra

$$A[s][f_1/s, \ldots, f_r/s] = A[s, y_1, \ldots, y_r]/(sy_i - f_i)$$

contains $i'(Z \cap \mathrm{Spec}(A))$ as the closed subscheme cut out by $y_1, \ldots, y_r$. Since $y_1, \ldots, y_r, sy_1 - f_1, \ldots, sy_r - f_r$ is a regular sequence in the polynomial ring $A[s, y_1, \ldots, y_r]$ we find that i' is a regular immersion. Some details omitted. As above we conclude that $Q = i'_*\mathcal{O}_{\mathbf{P}^1_Z}$ is a perfect object of $D(\mathcal{O}_W)$. All the other assumptions on F and Q in Lemma 0FE8 (and Lemma 0F9K) are immediately verified. $\square$